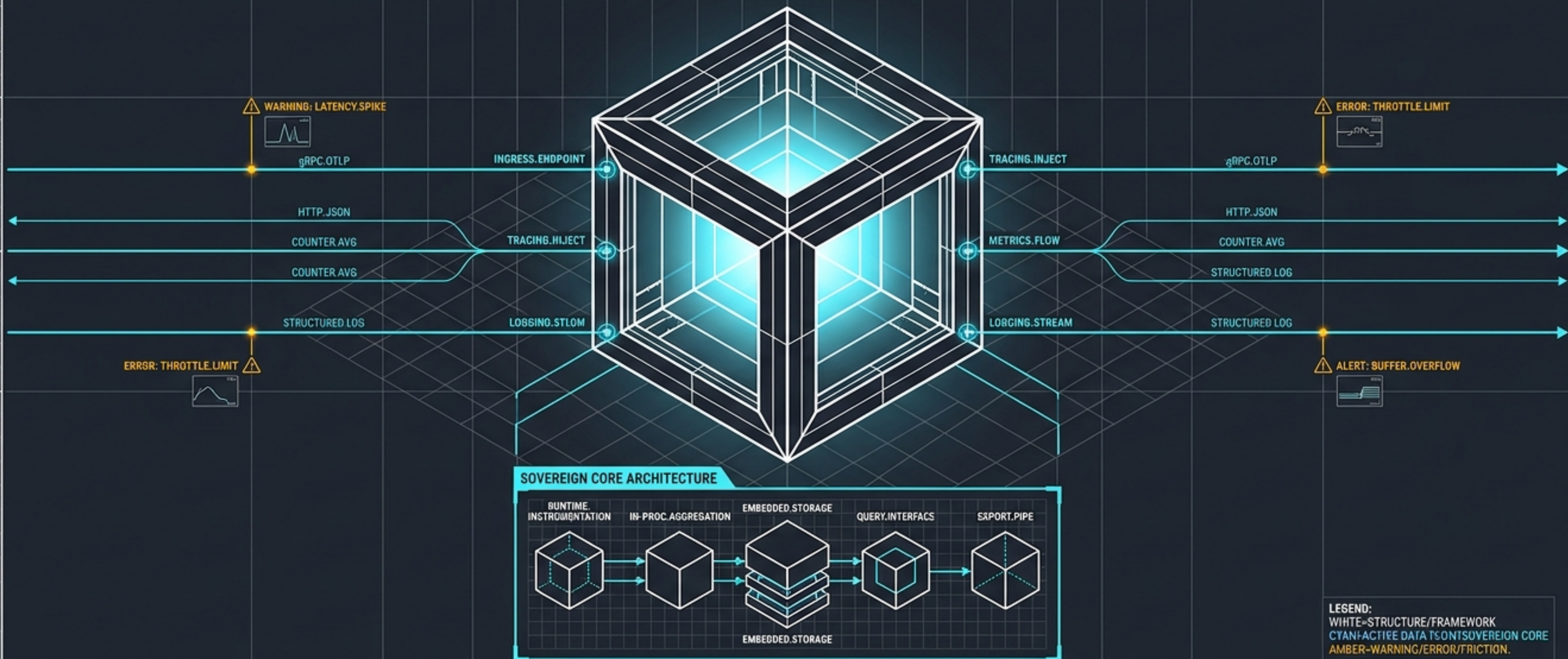


SOVEREIGN OBSERVABILITY.

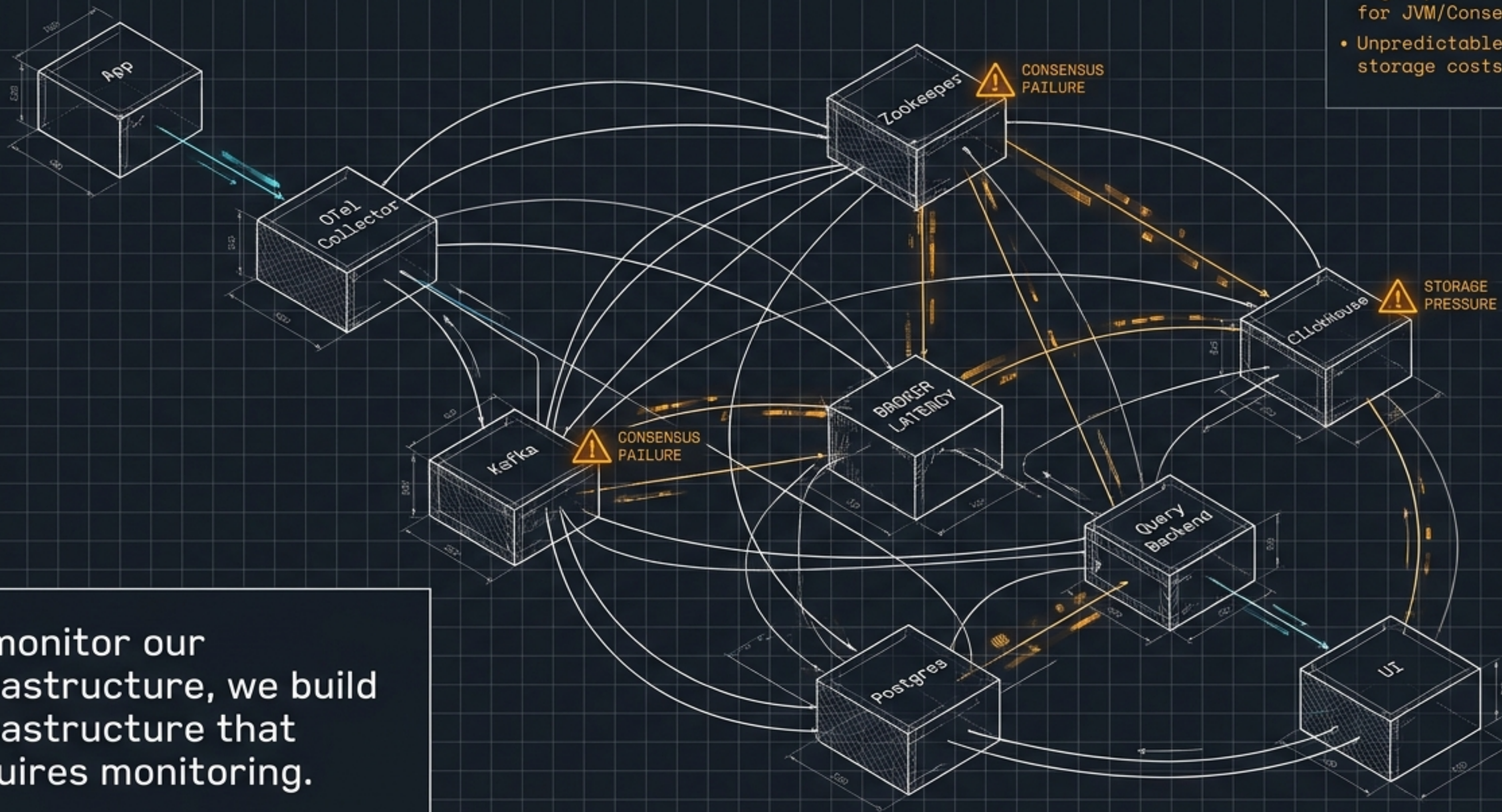
A Radical Blueprint for Single-Binary .NET 8/9 APM.



THE OBSERVABILITY PARADOX

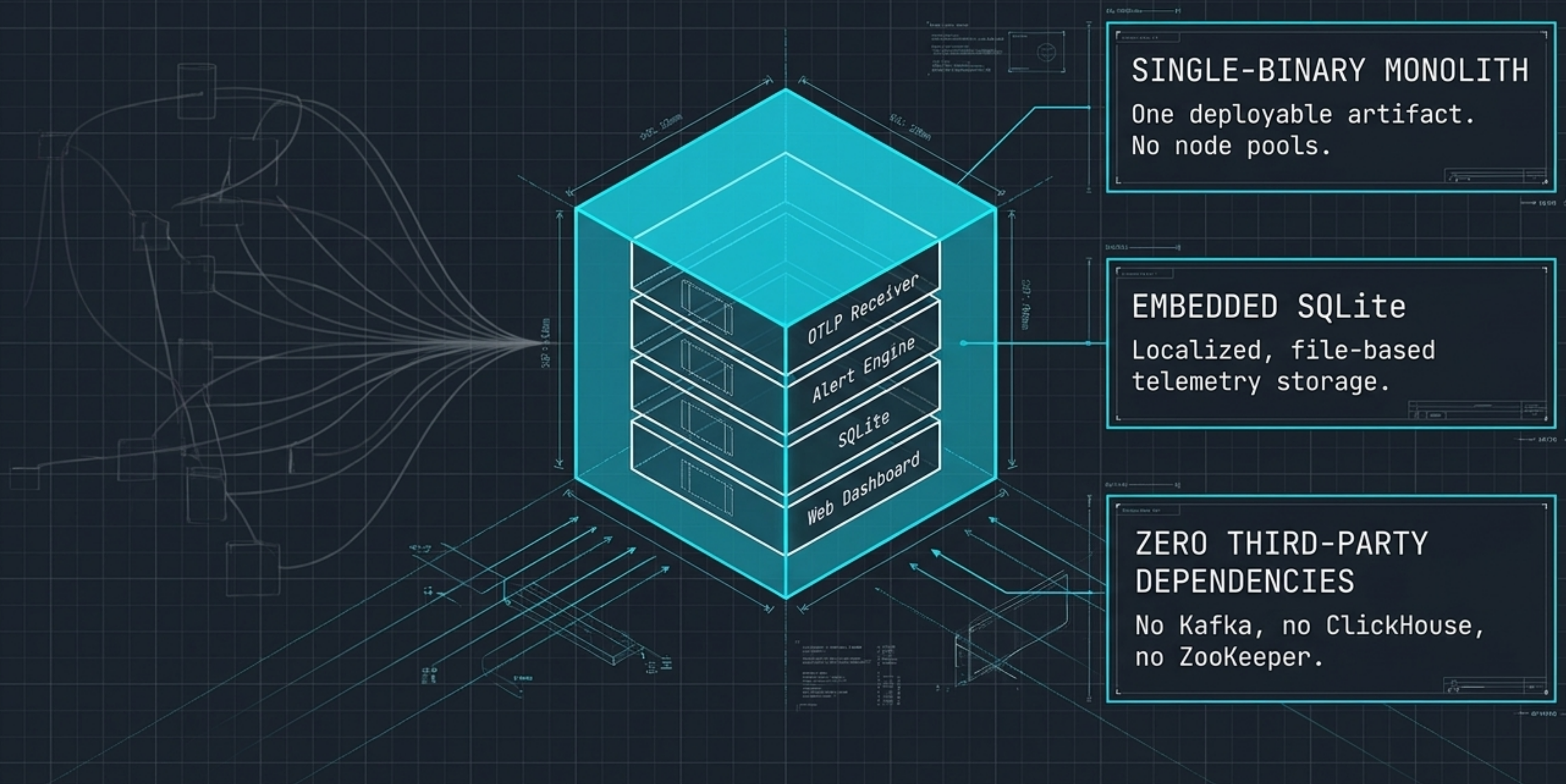
DIAGNOSTIC

- Latency injected by intermediaries
- High operational overhead for JVM/Consensus nodes
- Unpredictable stateful storage costs



To monitor our infrastructure, we build infrastructure that requires monitoring.

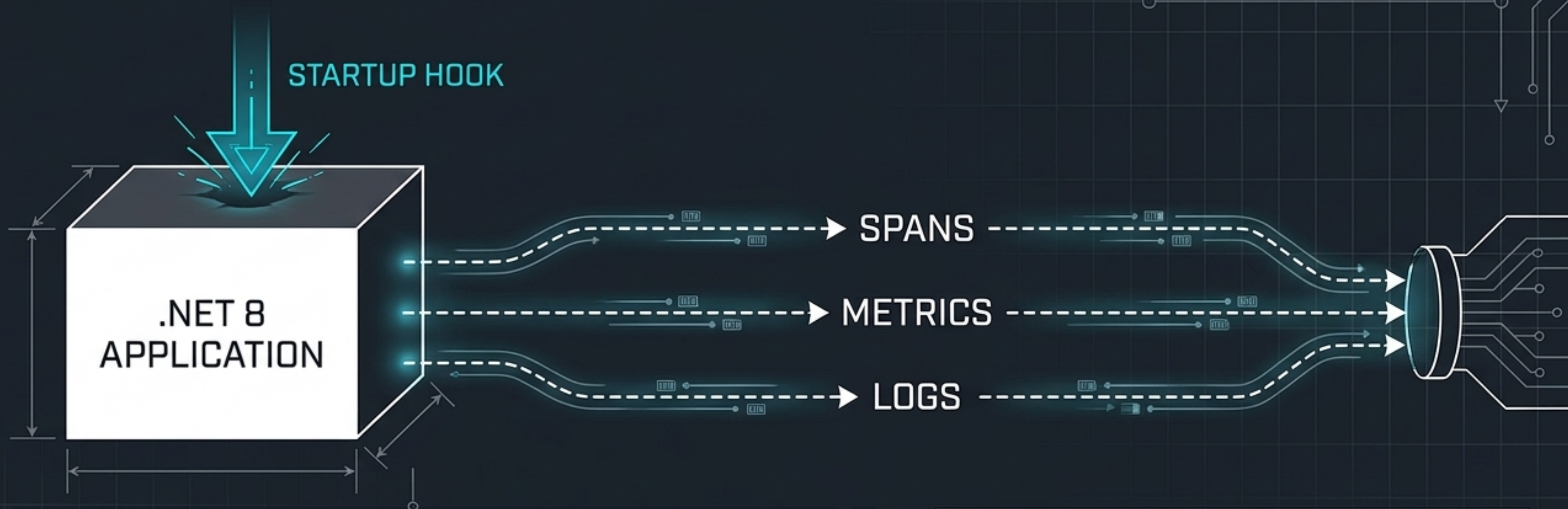
THE SOVEREIGN BLUEPRINT



ARCHITECTURAL DIAGNOSTICS: HEAVYWEIGHT VS. SOVEREIGN

	HEAVYWEIGHT APM	SOVEREIGN APM
Architecture	Multi-node Distributed (ClickHouse/Kafka)	Single-Binary Monolith
Dependencies	Postgres, ZooKeeper, ClickHouse	Zero (Embedded)
Data Ingestion	Multi-hop network latency	Direct in-process buffering
Deployment Time	Days/Weeks (Helm, PVCs)	< 5 Minutes
Ideal Target	Global Enterprise Clusters	Sovereign, Air-gapped, Self-Hosted

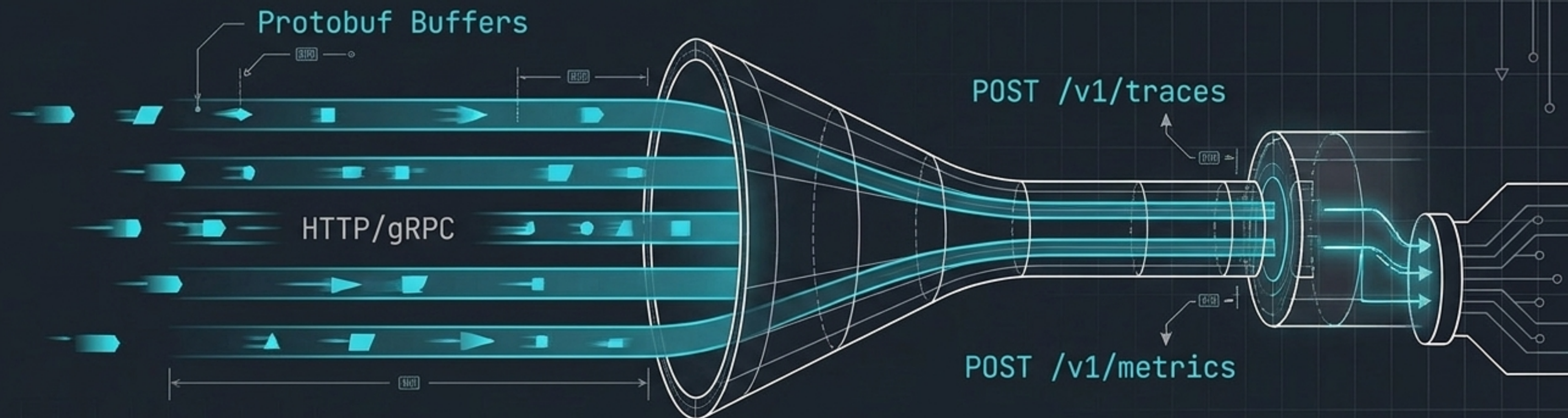
ZERO-CODE .NET 8/9 INSTRUMENTATION



```
export DOTNET_STARTUP_HOOKS=/opt/apm/ApmAgent.dll  
export OTEL_SERVICE_NAME=billing-service
```

- No source code changes required.
- The CLR Profiler intercepts the bootstrap process at runtime.
- Automatically captures structural traces, HTTP boundaries, and diagnostic metrics via standard environment variables.

NATIVE OTLP INGESTION (NO INTERMEDIARIES)



DIRECT & ASYNCHRONOUS

Non-blocking telemetry writes directly to the monolith.

HOST PROTECTION

Immediate HTTP acknowledgment responses protect the target application from network or database bottlenecks.

NATIVE COMPATIBILITY

100% OTLP compliant.
No standalone OTel Collector required.

THE STORAGE ENGINE: EMBEDDED SQLITE (WAL MODE)

WRITE-AHEAD LOGGING:

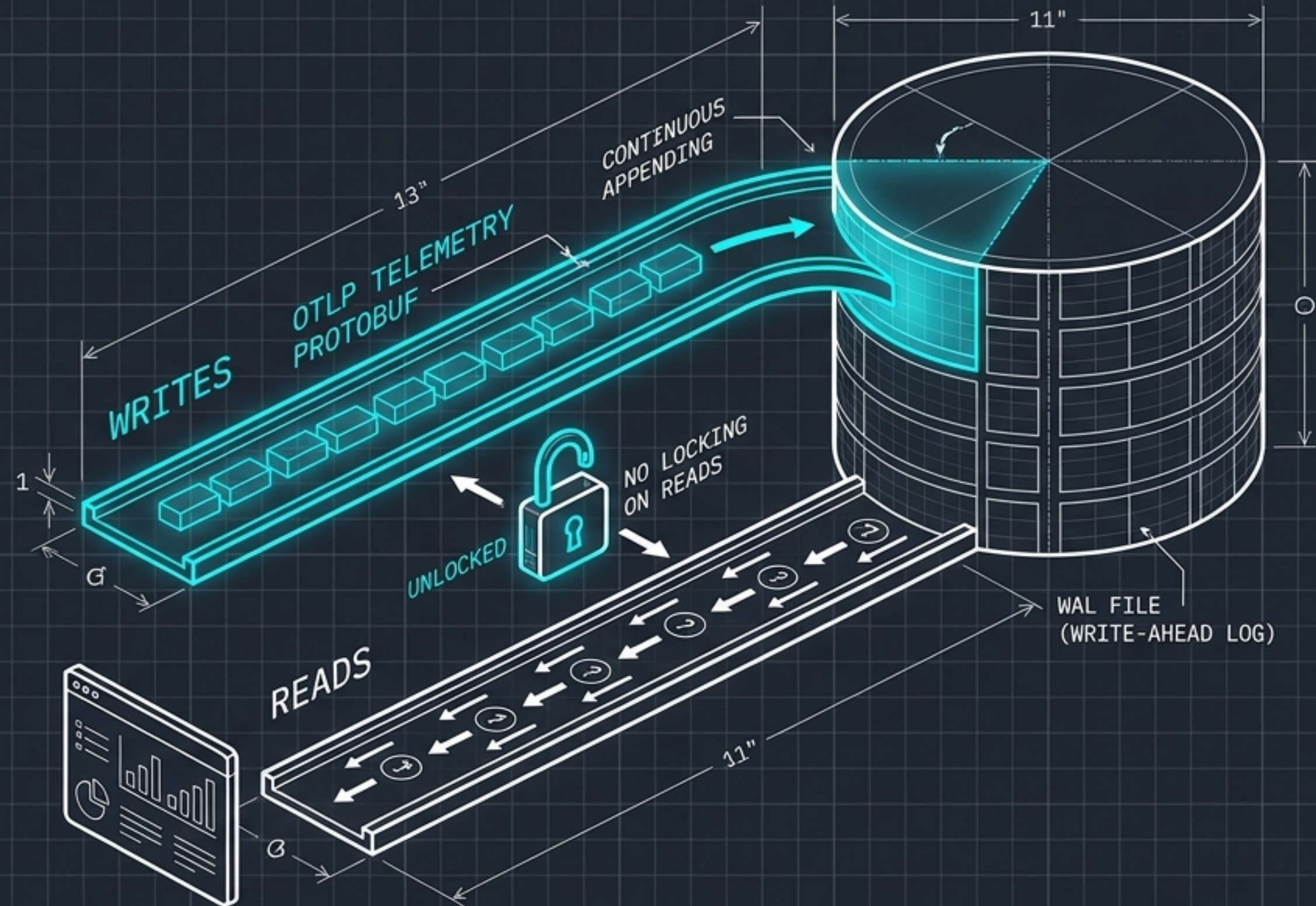
Solves concurrency by allowing continuous appending of telemetry data without locking the read database.

HIGH-PERFORMANCE INDEXING:

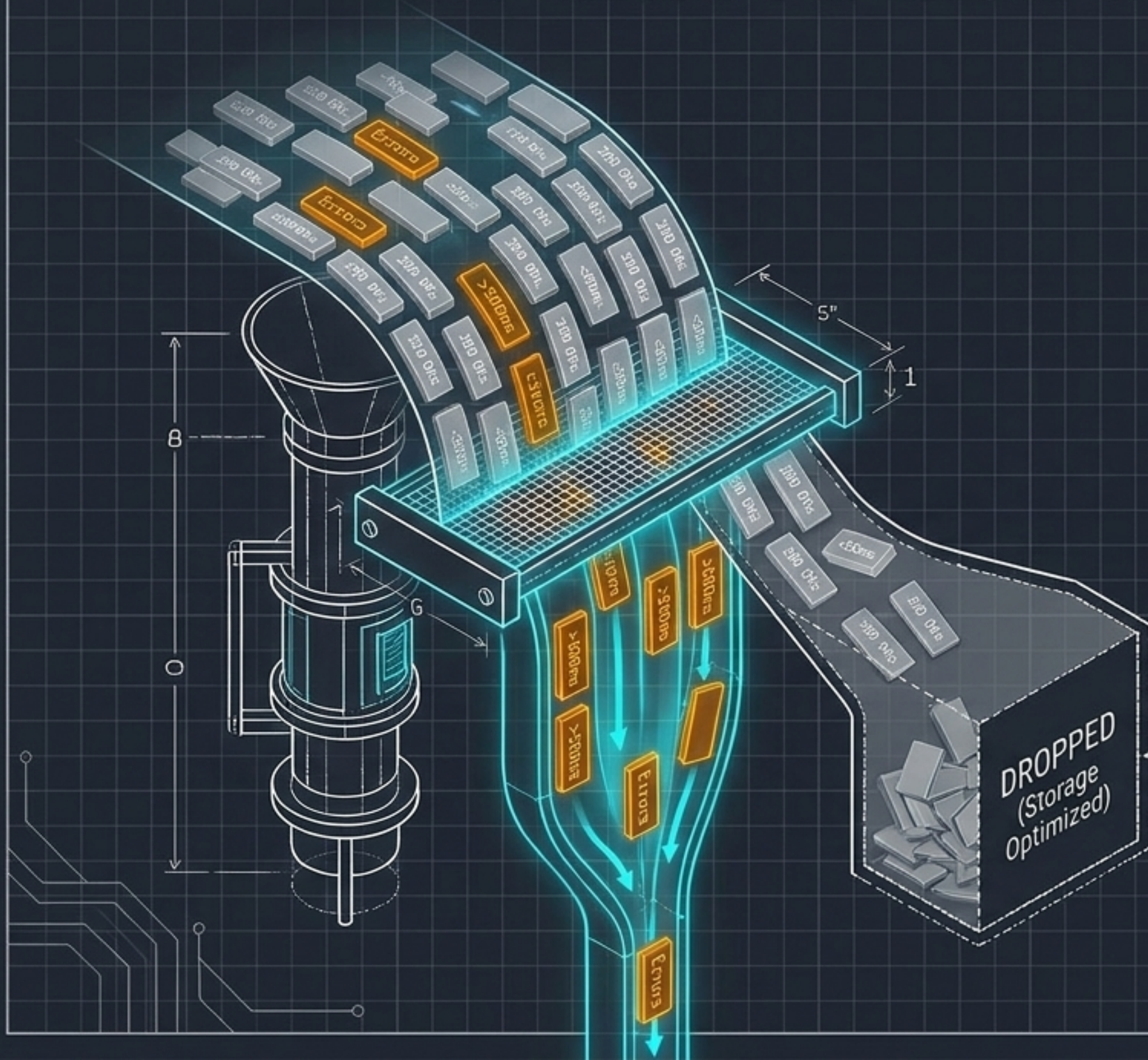
Columnar-style access patterns inside a highly optimized local file.

THE RESULT:

Enterprise-grade write throughput without the operational overhead of a distributed database cluster.



THE TRACE PIPELINE & TAIL SAMPLING



INTELLIGENT TAIL-BASED SAMPLING

The in-memory pipeline analyzes complete traces before committing to disk.

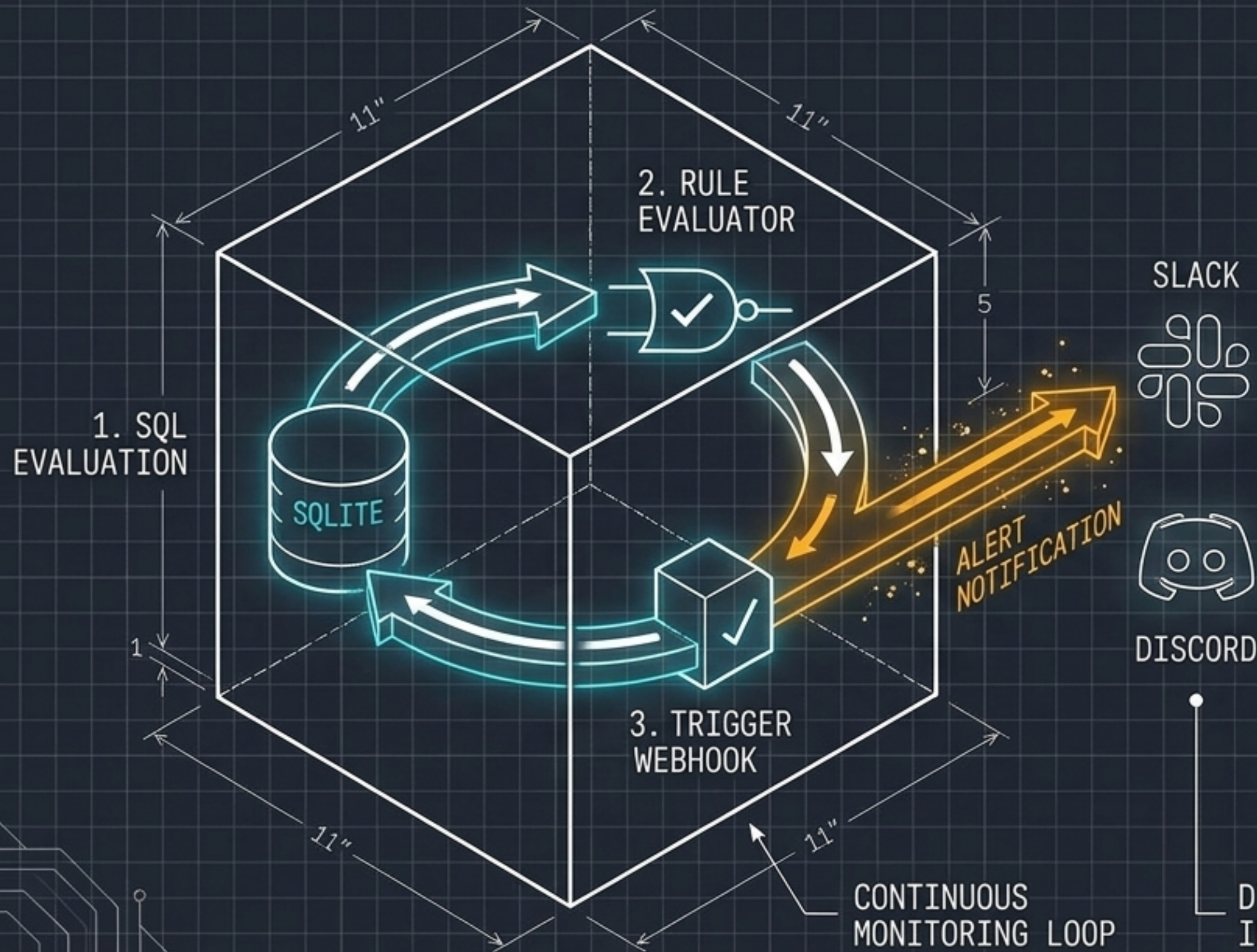
ZERO CRITICAL DATA LOSS

100% of exceptions, errors, and slow queries bypass the sampling filter.

STORAGE OPTIMIZATION

Routine, successful requests are dynamically dropped, optimizing local storage costs without sacrificing visibility.

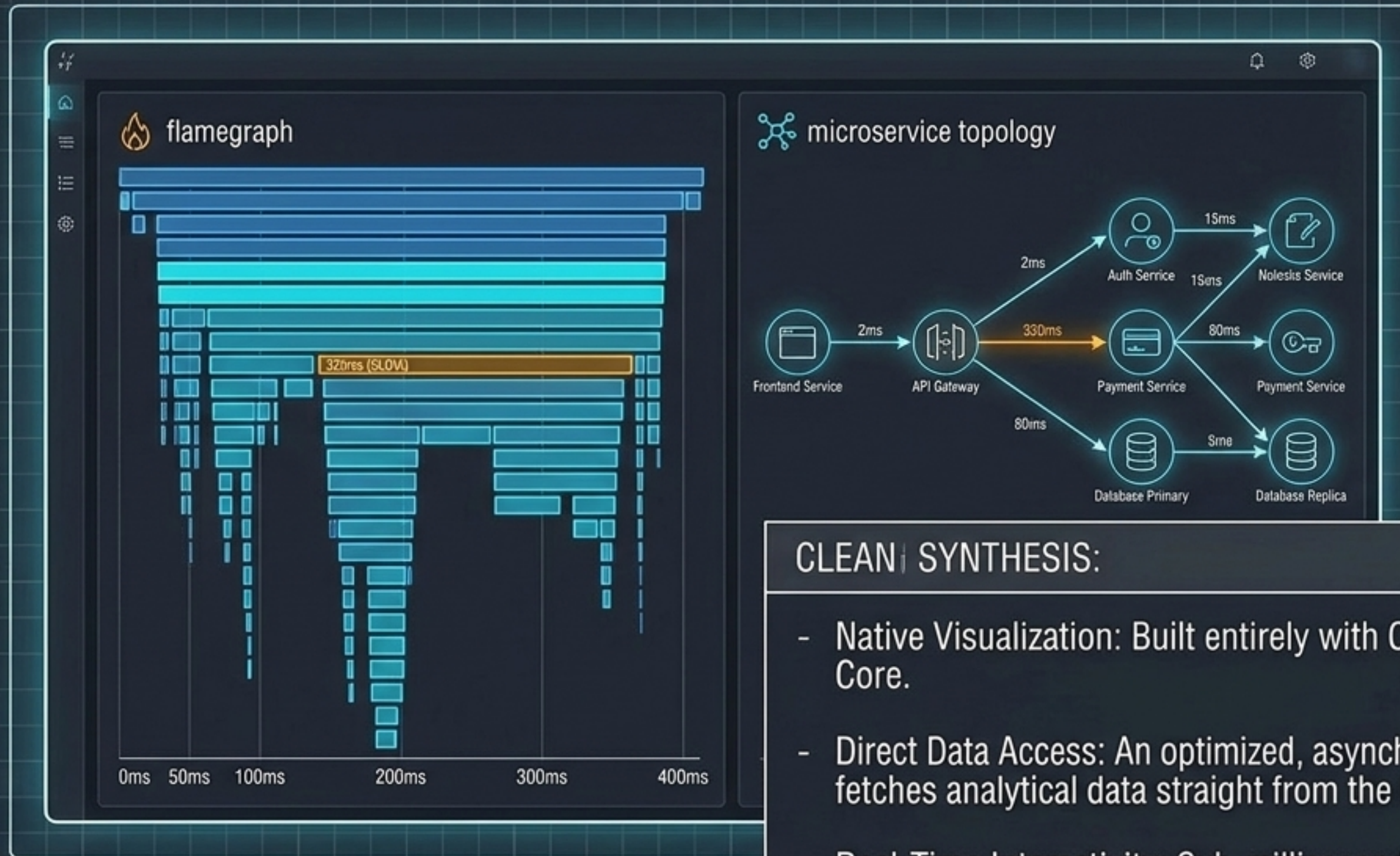
IN-PROCESS BACKGROUND ALERTING ENGINE



TECHNICAL BREAKDOWN:

- No separate Cron jobs. No external Alertmanager.
- A seamless, continuous background C# thread querying local metrics.
- Detects service degradation, error spikes, and latency breaches before end-users do.

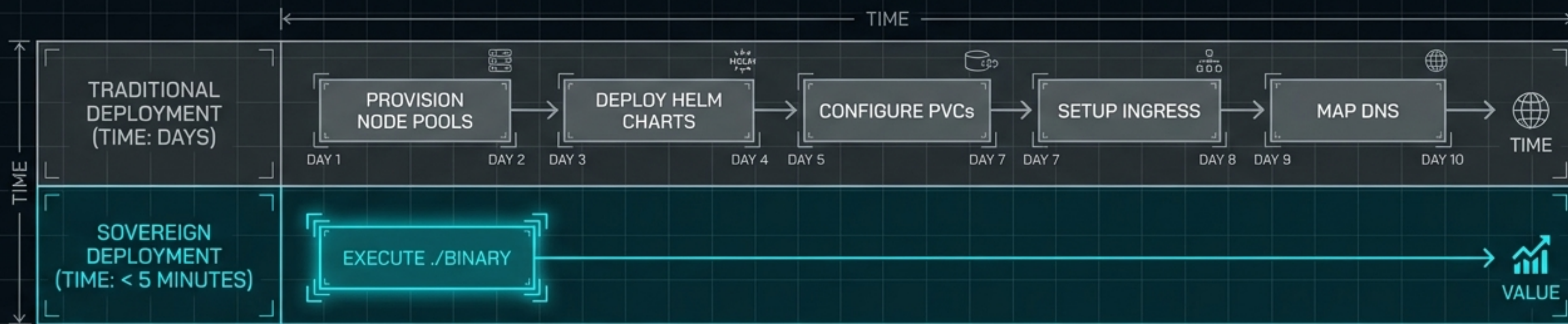
LOCAL WEB DASHBOARD UI



CLEAN | SYNTHESIS:

- Native Visualization: Built entirely with C# and ASP.NET Core.
- Direct Data Access: An optimized, asynchronous query engine fetches analytical data straight from the local SQLite engine.
- Real-Time Interactivity: Sub-millisecond rendering of application topology and transaction breakdowns.

DEPLOYMENT: FIVE MINUTES TO VALUE



```
docker run -d -p 4317:4317 -p 8080:8080 \ \
-v /var/data/observability:/data \ \
sovereign-apm:latest
```

FROM DOWNLOAD TO FIRST TRACE IN UNDER 5 MINUTES. NO INFRASTRUCTURE PROVISIONING REQUIRED. TRUE OPERATIONAL INDEPENDENCE.

One Binary. Infinite Visibility.

The raw power and expertise of the world's best observability platforms, distilled into a single, highly engineered, self-hosted engine.

- Zero code changes.
- Zero external databases.
- Complete operational sovereignty.

